



CHEMISTRY CH.11 — SOLUTIONS — FULL MIND MAP

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Topics in this chapter:

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2. Types of Mixtures — Homogeneous vs Heterogeneous
3. Separation Techniques
4. Evaporation, Boiling & Phase Changes
5. Electrolytes & Conduction of Electricity
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16. Mass Ratio in Molecules
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1. Solutions, Suspensions & Colloids — Definitions & Differences

- Solution: Homogeneous mixture of two or more substances — solute uniformly dissolved in solvent. Examples: salt water, sugar water, air, carbonated beverages. [Q1415, Q1418, Q1423, Q1432]
- True Solution: Solute particles $< 10^{-9}$ m (1 nm) in diameter; homogenous; does NOT scatter light. [Q1427]
- Colloid (Colloidal Solution): Heterogeneous mixture; particle size 1 nm to 1000 nm (10^{-9} to 10^{-6} m); particles do NOT settle down when left undisturbed; individual particles not visible to naked eye; Tyndall effect dikhata hai (colloid particles light ko scatter karte hain — is size ki wajah se, jo true solution se bade aur suspension se chhote hote hain). Milk ek colloidal solution hai jo Tyndall effect dikhata hai; starch solution bhi. [Q1411, Q1413, Q1417, Q1424, Q1431, Q1434, Q1430]
 - Types of colloids: Sol = Solid dispersed in liquid (e.g., Gold sol, Sulfur sol, Arsenious sulphide sol, paint); Gel = Liquid dispersed in solid (e.g., Cheese, butter, jellies); Emulsion = Liquid dispersed in liquid (e.g., Milk, face cream, mayonnaise); Foam = Gas dispersed in liquid (e.g., Whipped cream, soap lather, soap suds); Aerosol = Liquid/Solid dispersed in gas (e.g., Cloud, mist, fog). [Q1411, Q1431, Q1434]
- Suspension: Heterogeneous mixture; solid particles dispersed but NOT dissolved; particles $> 10^{-6}$ m; visible to naked eye; scatter light; settle over time. [Q1418, Q1424, Q1427]
 - Example: Mud in water. [Q1427]
- Precipitate: Chemical reaction ke dauraan solution mein bana insoluble substance. Example: $\text{AgNO}_3(\text{aq}) + \text{KCl}(\text{aq}) \rightarrow \text{AgCl} \downarrow (\text{precipitate}) + \text{KNO}_3(\text{aq})$. [Q1405]
- ⚠ Trap: Colloid cannot scatter light — ye statement galat hai. Colloid Tyndall effect dikhaate hain. [Q1417]
- ⚠ Trap: Colloid heterogeneous mixture hai, homogeneous nahi. [Q1417, Q1434]

2. Types of Mixtures — Homogeneous vs Heterogeneous

- Homogeneous: Uniform composition throughout; single phase. Examples: salt + water, sugar + water, air, alcohol

+ water. [Q1415, Q1429, Q1432]

- Heterogeneous: Non-uniform composition; visible separate phases/particles. Examples: sand + water, oil + water, colloids, suspensions. [Q1417, Q1424, Q1429, Q1432, Q1434]
- Mixture can be both homogeneous and heterogeneous. [Q1432]
- ⚠ Trap: Pure substance = NOT a mixture. [Q1432]

3. Separation Techniques

- Filtration: Separating suspended solid particles from liquid/gas using filters. Example: water from muddy water. [Q1426, Q1437]
- Evaporation: Liquid → gas at surface; used to separate volatile component from non-volatile solute. Example: salt from salt water. [Q1425, Q1426]
- Separating Funnel: Separating immiscible liquids (different densities, form separate layers). Example: petrol and water. [Q1429]
- Centrifugation (Churning): Using centrifugal force to separate components; e.g., cream from milk. [Q1426, Q1437]
- Sublimation: Solid → gas without liquid phase. Example: salt and camphor separated by sublimation. [Q1429]
- Chromatography: Separating dye from black ink. [Q1429]
- Decantation: Separating liquid from solid/immiscible liquids by removing top liquid layer. [Q1437]
- Sieving: Separating particles of different sizes. [Q1437]
- Diffusion: Movement of substance from high concentration to low concentration. Example: smell of perfume reaches nose. [Q1419]
- Condensation: Gas → liquid. Example: morning dew, foggy car windows. [Q1419, Q1425]
- ⚠ Trap: Filtration muddy water ke liye best hai — distillation ya crystallization basic separation ke liye overkill hain. [Q1437]

4. Evaporation, Boiling & Phase Changes

- Evaporation: Surface phenomenon; liquid → gas; occurs at ALL temperatures (not just boiling point). Factors affecting rate:
 - Temperature ↑ → rate ↑
 - Surface area ↑ → rate ↑
 - Wind speed ↑ → rate ↑
 - Humidity ↑ → rate ↓ [Q1420, Q1425, Q1435, Q1436]
- Boiling: Bulk phenomenon; occurs at boiling point (100°C for water at sea level). [Q1435, Q1436]
- Freezing: Liquid → solid. [Q1436]
- Condensation: Gas → liquid; reverse of vaporization. [Q1419, Q1425, Q1436]
- Sublimation: Solid → gas; no liquid intermediate. Examples: Camphor, Iodine, Dry ice (solid CO₂). [Q1422, Q1426]
- ⚠ Trap: Evaporation surface phenomenon hai — boiling bulk phenomenon hai. [Q1435]
- ⚠ Trap: Same quantity of liquid ke liye evaporation sabse slowest hota hai freezing, boiling, condensation ke comparison mein. [Q1436]
- Physical state of water at 250°C: Gas (above boiling point 100°C).
 - Below 0°C → Solid (ice)
 - 0°C to 100°C → Liquid
 - Above 100°C → Gas [Q1421]

5. Electrolytes & Conduction of Electricity

- Electrolyte: Substance that conducts electricity when dissolved in water or molten state; dissociates into ions. [Q1418]
- Ionic compounds (NaCl , CaCl_2 , CuSO_4) water mein dissociate hoke ions banate hain — isliye electricity conduct karte hain. [Q1404]
- Octane (C_8H_{18}): Covalent compound; water mein dissociate nahi hota; NO ions $\rightarrow$ bulb does NOT glow. [Q1404]
- Examples of electrolytes: NaCl(aq) , $\text{CaCl}_2\text{(aq)}$, $\text{CuSO}_4\text{(aq)}$. [Q1404]
- ⚠ Trap: Covalent liquids like octane electrolyte nahi hote — bulb nahi jalega. [Q1404]

6. Important Reactions & Precipitates

- $\text{AlCl}_3 + \text{NH}_4\text{OH} \rightarrow \text{Al(OH)}_3\downarrow$ (white ppt) + NH_4Cl . Aluminium hydroxide = white precipitate; used for heartburn, sour stomach, peptic ulcer pain. [Q1402]
- Limestone (CaCO_3) + dilute $\text{HCl} \rightarrow \text{CO}_2\uparrow + \text{CaCl}_2 + \text{H}_2\text{O}$. [Q1407]
 - $\text{CO}_2 + \text{Lime water (Ca(OH)}_2) \rightarrow \text{CaCO}_3\downarrow$ (white ppt) + H_2O . [Q1407]
 - Excess $\text{CO}_2 + \text{CaCO}_3 + \text{H}_2\text{O} \rightarrow \text{Ca(HCO}_3)_2$ (soluble). [Q1407]
- $\text{Pb(NO}_3)_2 + 2\text{NaI} \rightarrow \text{PbI}_2\downarrow$ (yellow ppt) + 2NaNO_3 . Lead iodide = yellow crystalline solid; insoluble in water; denser than water; used in photography, solar cells. [Q1410]
- $\text{Zn} + \text{FeCl}_3 \rightarrow \text{ZnCl}_2$ (colourless/white, highly soluble). Yellow colour of ferric chloride disappears. [Q1406]
- $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ (brisk effervescence). [Q1251]

7. pH, Indicators & Bases

- pH scale:
 - Acidic: 0 to <7
 - Neutral: 7
 - Basic (Alkaline): >7 to 14 [Q1416]
- Alkaline solutions: NaOH , Mg(OH)_2 , Ca(OH)_2 , KOH — $\text{pH} > 7$. [Q1416]
- Aluminium hydroxide (Al(OH)_3): Amphoteric — both acidic and basic properties; isliye alkaline nahi kehlaata. [Q1416]
- Phenolphthalein: Pink in basic solution; colourless in acidic. [Q1403]
- Turmeric: Red in base. [Q1403]
- Red cabbage: Red in acid; green in base. [Q1403]
- Litmus: Acid $\rightarrow$ blue litmus red; Base $\rightarrow$ red litmus blue. [Q1403]
- Methyl orange: Red in acid; yellow in base. [Q1403]
- $\text{Mg} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$ (basic) — phenolphthalein pink kar deta hai. [Q1403]
- ⚠ Trap: Al(OH)_3 amphoteric hai — alkaline nahi. [Q1416]

8. Solubility & Concentration

- Solubility: Amount of solute present in saturated solution at a particular temperature — ability of solute to dissolve in solvent. [Q1428]

- Saturated solution: Dissolved as much solute as it is capable of dissolving. [Q1424]
- Unsaturated solution: Dissolved less solute than it is capable of dissolving at given T and P. [Q1424]
- Concentrated solution: Relatively large amount of solute dissolved in given amount of solvent. [Q1423]
- Isotonic solution: Concentration of solutes same as another solution or cell interior. [Q1423]
- Aqueous solution (aq): Solute dissolved in water. [Q1423]
- Brine: Aqueous solution of sodium chloride (NaCl). [Q1412]
- Chlor-alkali process: Electrolysis of brine $\rightarrow 2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{Cl}_2 + 2\text{NaOH}$ (Castner-Kellner cell). [Q1408]
 - Cl_2 at anode; H_2 and NaOH at cathode. [Q1408]

9. Density & Relative Density

- Density (ρ): Mass per unit volume; $\rho = m/V$. SI unit: kg/m^3 ; dimensional formula: $[\text{ML}^{-3}\text{T}^0]$. [Q1433]
- Relative density: $\rho_{\text{substance}} / \rho_{\text{water}}$ (reference material). Dimensional formula: $[\text{M}^0\text{L}^0\text{T}^0]$ (dimensionless). [Q1433]
- Viscosity: Measure of resistive flow of fluid under gravity; internal friction of moving fluid. Example: Honey (due to hydrogen bonding). [Q1428]

10. Physical States & Key Properties

- Solids: Molecules closely bound; examples: Iron, brick, coin. [Q1421]
- Liquids: Molecular forces weaker than solids; examples: Juice, cold drinks. [Q1421]
- Gases: Molecular forces very weak; examples: Natural gas, O_2 , CO_2 . [Q1421]
- Plasma: Liquid portion of blood; helps in blood clotting. (Biological context) [Q1421]
- CuCl_2 solution: Blue-green colour; concentrated = dark brown; dilute = green $\rightarrow$ blue (Cl^- replaced by water molecules). [Q1409]
- Iodine sublimation vapours: Purple colour. [Q1422]
- Ammonia (NH_3): Colourless, sharp suffocating odour; N:H mass ratio = 14:3; ~80% used as fertilizer; also refrigerant, water purification, plastics, explosives. [Q1414]
- Chlorine (Cl_2): Greenish-yellow, toxic, corrosive, irritating to eyes/respiratory system. [Q1340]

11. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Colloid cannot scatter light — ye statement galat hai. Colloid Tyndall effect dikhaate hain. [Q1417]
- ⚠ Trap: Colloid = heterogeneous mixture, NOT homogeneous. [Q1434]
- ⚠ Trap: $\text{Al}(\text{OH})_3$ amphoteric hai — alkaline nahi. [Q1416]
- ⚠ Trap: Evaporation sirf boiling point pe nahi — sabhi temperatures pe hota hai. [Q1420]
- ⚠ Trap: Evaporation = surface phenomenon; Boiling = bulk phenomenon. [Q1435]
- ⚠ Trap: Same quantity ke liye evaporation sabse slowest hai. [Q1436]
- ⚠ Trap: Octane (covalent) electrolyte nahi — bulb nahi jalega. [Q1404]
- ⚠ Trap: Pure substance = NOT a mixture. [Q1432]
- ⚠ Trap: True solution particles $< 10^{-9}$ m; Colloid 1–1000 nm; Suspension $> 10^{-6}$ m. [Q1427]
- ⚠ Trap: Separating funnel = immiscible liquids ke liye; miscible liquids ke liye nahi. [Q1429]
- ⚠ Trap: Aqueous (aq) = solution in water, kerosene nahi. [Q1423]

12. Ideal Gas Law ($PV = nRT$) — Vapour Phase Calculations

- Ideal Gas Equation: $PV = nRT$. [Q1438, Q1342]
 - P = Pressure (Pa), V = Volume (m^3 ya dm^3), n = moles, $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$, T = Temperature (K). [Q1438]
- Question trick: Flask mein zyada water daala gaya hai, lekin equilibrium pe sirf vapour phase ka hi calculate karna hai — liquid phase ignore karo. [Q1438]
- Calculation:
 - $n = PV/RT = (3170 \text{ Pa} \times 1 \text{ dm}^3) / (8.314 \times 300 \text{ K}) = 1.27 \times 10^{-3} \text{ mol}$. [Q1438]
- Exam tip: Agar flask ka volume diya hai aur vapour pressure diya hai, toh liquid ka volume matter nahi karta — sirf flask ka volume vapour ka volume hota hai. [Q1438]

13. Dalton's Law of Partial Pressures

- Dalton's Law: Total pressure = sum of partial pressures; partial pressure $\propto$ mole fraction. [Q1439]
- Mole fraction = moles of component / total moles. [Q1439]
- Question: Equal masses of ethane (C_2H_6 , $M=30$) and hydrogen (H_2 , $M=2$) mixed. H_2 ka pressure fraction nikaalo.
- Trick:
 - Let mass of each = x g.
 - Moles of ethane = $x/30$; moles of H_2 = $x/2$.
 - Mole fraction of $\text{H}_2 = (x/2) / (x/2 + x/30) = (1/2) / (1/2 + 1/30) = 15/16$. [Q1439]
- Shortcut: Equal mass mix karne pe, lighter gas (H_2) ka mole fraction hamesha zyada hota hai. [Q1439]

14. Stoichiometry — Mass-Mass Calculations

- Reaction: $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4\downarrow + 2\text{HCl}$. [Q1440]
- Molar masses:
 - $\text{BaCl}_2 = 137 + (35.5 \times 2) = 208 \text{ g/mol}$.
 - $\text{BaSO}_4 = 137 + 32 + (16 \times 4) = 233 \text{ g/mol}$. [Q1440]
- Logic: $208 \text{ g BaCl}_2 \rightarrow 233 \text{ g BaSO}_4$.
 - $1 \text{ g BaCl}_2 \rightarrow 233/208 \text{ g BaSO}_4$.
 - $2.08 \text{ g BaCl}_2 \rightarrow (233/208) \times 2.08 = 2.33 \text{ g BaSO}_4$. [Q1440]
- Exam trick: $2.08 = 208/100$, toh direct $233/100 = 2.33$ aa jaata hai — calculation fast karo. [Q1440]

15. Percentage Composition

- Formula: % of element = (atomic mass of element / molecular mass of compound) $\times 100$. [Q1441]
- CO_2 ka calculation:
 - $\text{C} = 12$, $\text{O} = 16 \times 2 = 32$.
 - Total = $12 + 32 = 44$.
 - % C = $(12/44) \times 100 = 27.27\% \approx 27.3\%$. [Q1441]
- Shortcut: $12/44 = 3/11 \approx 0.2727 \rightarrow 27.27\%$. [Q1441]

16. Mass Ratio in Molecules

- Water molecule (H_2O):
 - 2 H atoms = $2 \times 1 = 2$ g.
 - 1 O atom = 16 g.
 - Mass ratio H : O = $2 : 16 = 1 : 8$. [Q1442]
- ⚠ Trap: Atom ratio H:O = 2:1 hai, lekin mass ratio = 1:8 hai — dono alag hain. [Q1442, Q1388]

17. Mass by Mass Percentage (m/m%)

- Formula: $\text{m/m\%} = (\text{mass of solute} / \text{mass of solution}) \times 100$. [Q1443]
- Question: 250 g of 5% glucose solution — kitna glucose chahiye?
- Calculation: Mass of glucose = $(250 \times 5) / 100 = 12.5$ g. [Q1443]
- Trick: 5% = $1/20$, toh $250/20 = 12.5$ directly. [Q1443]

18. ⚠ Numerical Traps / Key Differences Summary

- ⚠ Trap: Vapour pressure calculation mein liquid ka volume ignore karo — sirf container ka volume vapour ka volume hota hai. [Q1438]
- ⚠ Trap: Equal mass mix karne pe mole fraction $\neq$ mass fraction — mole fraction mein lighter gas zyada dominate karti hai. [Q1439]
- ⚠ Trap: BaCl_2 (208) aur BaSO_4 (233) ka molar mass yaad rakho — $208 \rightarrow 233$ direct ratio se calculation fast hoti hai. [Q1440]
- ⚠ Trap: % composition mein molecular mass se divide karna hai, atomic mass se nahi. [Q1441]
- ⚠ Trap: H_2O mein atom ratio (2:1) aur mass ratio (1:8) alag hain — question kya puch raha hai dhyan se padho. [Q1442]
- ⚠ Trap: m/m% mein solution ka mass denominator mein aata hai, sirf solvent ka nahi. [Q1443]